University of Ottawa
School of Electrical Engineering and Computer Science

# Robust Tabular Classification under Noisy Labels: Benchmarking and Lightweight Cleaning

CSI6900 Research Project Report

Ovais Azeem
Student number: 300112311

Supervisor: Professor Thomas Tran

October 3, 2026

# Abstract

Label errors can impair tabular classifiers, but detecting uncertain samples does not necessarily identify incorrect labels. This project benchmarks logistic regression, random forest, and histogram gradient boosting on four datasets with training-only symmetric and cyclic class-dependent corruption. Four noise rates, four seeds, and four out-of-fold detectors produce 3,840 evaluation rows. Confidence, disagreement, margin, and a weighted combined score guide removal or relabeling. Confidence-based removal improves mean test accuracy by 4.27 percentage points over noisy training; combined-score removal improves it by 3.68 points. Confidence also provides the highest average detection F1. Margin-only cleaning is usually harmful. The combined score does not outperform confidence overall and largely preserves its binary ranking. Dataset-level inference with four datasets cannot establish significance at 0.05. An audited, containerized release preserves results, models, and figures. The findings support simple confidence filtering within this controlled, known-noise-budget benchmark, not universal superiority under real annotation noise.

# Contents

**Abstract** **i**

**1 Introduction** **1**

1.1 Motivation . . . . . 1

1.2 Research Questions . . . . . 1

1.3 Contributions and Scope . . . . . 2

1.4 Organization . . . . . 2

**2 Related Work** **3**

2.1 Noise Models and Robust Learning . . . . . 3

2.2 Evidence Specific to Tabular Data . . . . . 3

2.3 Confidence Is Not Confident Learning . . . . . 4

2.4 Ground Truth and Statistical Evidence . . . . . 4

2.5 Position of This Project . . . . . 4

**3 Proposed Approach** **5**

3.1 Notation and Experimental Isolation . . . . . 5

3.2 Controlled Label Corruption . . . . . 5

3.3 Training-Only Out-of-Fold Probabilities . . . . . 6

3.4 Suspiciousness Scores . . . . . 6

3.5 Selection and Cleaning . . . . . 7

3.6 Experiment Algorithm and Cost . . . . . 7

**4 Experimental Setup** **9**

4.1 Datasets and Preprocessing . . . . . 9

4.2 Models and Fixed Parameters . . . . . 10

4.3 Matrix and Paired Conditions . . . . . 10

4.4 Evaluation Metrics . . . . . 11

4.5 Aggregation and Inference . . . . . 11

4.6 Execution and Verification . . . . . 12

**5 Results and Analysis** **13**
5.1 Aggregate Classification Performance . . . . . 13
5.2 RQ1: Sensitivity to Noise . . . . . 13
5.3 RQ2: Detection of Injected Errors . . . . . 14
5.4 RQ3: Effects of Cleaning and Relabeling . . . . . 18
5.5 Threats to Validity . . . . . 20

**6 Conclusion and Future Work** **22**
6.1 Answers to the Research Questions . . . . . 22
6.2 Contribution Supported by the Evidence . . . . . 22
6.3 Priorities for Future Work . . . . . 23

**References** **24**

**A Reproducibility and Supplementary Diagnostics** **26**
A.1 Immutable Evidence . . . . . 26
A.2 Artifact Map . . . . . 26
A.3 Build and Reproduction Commands . . . . . 27
A.4 Preparation and Review . . . . . 27
A.5 Supplementary Macro F1 and Curves . . . . . 28

# List of Tables


4.1 Dataset dimensions and exact split sizes. . . . . 9
4.2 Estimator settings. Unlisted settings retain scikit-learn 1.6.1 defaults. . . . . 10

5.1 Overall test performance across the balanced matrix. . . . . 13
5.2 Accuracy loss from clean to noisy training, in percentage points. . . . . 14
5.3 Mean test accuracy by dataset and classifier. . . . . 17
5.4 Injected-label-error detection, averaged over 384 combinations per detector. . . . . 17
5.5 Paired accuracy effects and exploratory dataset-level inference. . . . . 19
5.6 Combined scoring minus each component detector under the same action. . . . . 19
5.7 Illustrative saved-model setting: Dry Bean/RF/symmetric 20%/seed 42. . . . . 20

# List of Figures


5.1 Accuracy under symmetric corruption. Each point averages three models and four seeds within a dataset. Selected methods are shown for legibility; all methods appear in Table 5.1. Lines connect the four evaluated rates and do not represent fitted continuous trends. . . . . 15

5.2 Accuracy under cyclic class-dependent corruption, aggregated as in Fig. 5.1. High-rate cyclic corruption is particularly damaging on the multiclass tasks. . . . . 16

5.3 Detection F1 by nominal rate and corruption mechanism. Each point averages the four datasets, three models, and four seeds. No test labels are used to compute a suspiciousness score. . . . . 17

5.4 Paired accuracy gains over noisy training, in percentage points. Every cell averages 24 settings and four seeds for one dataset. Numeric annotations preserve interpretation without color. . . . . 18

5.5 Test confusion matrix for the illustrative noisy RF classifier. Rows are reference classes and columns predicted classes. The test set has 2,709 examples and was not corrupted. . . . . 21

A.1 Macro F1 under cyclic class-dependent corruption. Each point averages three models and four seeds within a dataset. The plotted methods and rates match the main accuracy comparison. . . . . 28

A.2 Micro-averaged ROC curve for the noisy Dry Bean RF classifier, symmetric 20% corruption and seed 42. Micro ROC AUC is approximately 0.9878; the main metric for this model is macro one-vs-rest AUROC, approximately 0.9874. . . . . 29

A.3 Micro-averaged precision–recall curve for the same noisy classifier. Trapezoidal PR AUC is approximately 0.9558. This is not average precision. . . . . 30

# Chapter 1

# Introduction

## 1.1 Motivation

Supervised classification learns relationships between observations and their labels. A label is often treated as a reliable target even when it was produced through an imperfect annotation process. Ambiguous definitions, transcription mistakes, incomplete evidence, and disagreement between annotators can create incorrect targets. The survey by Frénay and Verleysen distinguishes label noise from other data defects and reviews its consequences for classification and the methods used to address it [1].

This project focuses on numeric tabular representations. Such representations include measured chemical properties, image-derived morphology, and flattened pixel intensities. They share a matrix interface, but their class structure and difficulty differ. A method that works on one should not be assumed to work on all. Nazaretyan, Leser, and Kircher's tabular benchmark reports that no evaluated label-error identification method dominates all studied settings [2]. The practical question is therefore not merely whether a method can flag unusual observations, but whether its flags are useful for improving a particular downstream learner.

Two tasks must be separated. *Detection* identifies training rows whose observed labels are suspect. *Intervention* removes or changes those labels and then retrains a classifier. A high detection score does not guarantee that intervention improves classification. Removal can discard informative boundary examples, and relabeling can reinforce an incorrect prediction. This distinction motivates evaluating both tasks against separate targets: known injected corruptions for detection, and an untouched test set for downstream prediction.

## 1.2 Research Questions

The study addresses three questions from the project proposal:

1. **RQ1:** How does controlled training-label noise affect common tabular classifiers across datasets, corruption mechanisms, and noise rates?

2. **RQ2:** How accurately do lightweight out-of-fold confidence, disagreement, margin, and combined scores identify the injected errors?

3. **RQ3:** Does removing or relabeling the selected examples improve test accuracy, macro F1,

and AUROC relative to training on noisy labels?

The combined score is a specific hypothesis: assigning weights to several signals might produce a better ordering of suspicious examples than any component alone. It is evaluated against its individual components under an identical selection budget. Its implementation is not, by itself, evidence of novelty or superiority.

## 1.3 Contributions and Scope

The principal contribution is a reproducible controlled benchmark connecting label-error detection to downstream classification. It includes four datasets, three model families, two corruption mechanisms, four corruption rates, four random seeds, four detectors, and two cleaning actions. The study also provides a direct component comparison, an analytical explanation of the combined score's binary behavior, and an engineering release with original training provenance, fitted-model examples, generated figures, and checksum verification.

The empirical contribution includes a negative result: the weighted combination does not improve average performance over confidence alone. Reporting that outcome is essential to answering the proposed research question. The results support a limited practical conclusion about simple filtering under the conditions tested, not a claim that the score is a new state-of-the-art method.

The scope is deliberately controlled. Only training labels are corrupted; features are unchanged. The original dataset labels serve as reference labels, not independently certified ground truth. Selection uses the known injected noise fraction. No categorical-data pipeline, unknown-noise estimation, instance-dependent corruption, hyperparameter search, or neural model is evaluated. These limits delimit the conclusions rather than being hidden implementation details.

## 1.4 Organization

Chapter 2 situates the work within noisy-label learning and tabular evaluation. Chapter 3 defines the corruption, detection, and cleaning procedures. Chapter 4 specifies datasets, estimators, metrics, and statistical aggregation. Chapter 5 analyzes the results and failure cases. Chapter 6 concludes and identifies follow-up experiments. The appendix documents reproducibility and provides illustrative diagnostic curves.

# Chapter 2

# Related Work

## 2.1 Noise Models and Robust Learning

Frénay and Verleysen organize label-noise research around the assumptions that connect the observed and true labels [1]. That distinction matters because a uniform random flip is not equivalent to systematic confusion between particular classes or feature-dependent annotation errors. The present benchmark implements only uniform alternative-label flips and fixed cyclic transitions. Its controlled corruption mask makes detection measurable, but does not make either mechanism a realistic model of all annotation processes.

Li and Zhu review deep-learning approaches to noisy-label image classification and examine noise patterns informed by real-world data [3]. The current work occupies the narrower sample-selection and label-replacement part of this landscape. It does not adapt the training loss or architecture. This restriction allows the same cleaning interface to be used with linear, bagged-tree, and boosted-tree estimators, but leaves stronger robust-training approaches outside the experimental comparison.

## 2.2 Evidence Specific to Tabular Data

Nazaretyan et al. survey 34 approaches and benchmark 20 methods for identifying mislabeled tabular samples [2]. Their evaluation includes controlled corruption and a genomic dataset with known errors. The breadth of that study is greater than the present project. This project's role is a small, auditable comparison of inexpensive probability-based scores and the consequences of acting on their rankings. Its detection values should not be compared directly with theirs: datasets, noise processes, selection budgets, and learners are different.

Yang et al. investigate whether noisy-label methods associated with computer vision can help deep learning on electronic health records, and report beneficial adaptations in their evaluated settings [4]. Their work motivates testing rather than assuming transfer across domains. This project uses neither their EHR data nor their deep networks; its clinical dataset is a small image-derived diagnostic table. It therefore cannot validate clinical deployment or reproduce their domain-specific findings.

Eisenbürger et al. study label noise in gradient-boosted decision trees, including detection, relabeling, and early stopping [5]. They motivate including a boosted-tree learner, but this benchmark uses scikit-learn's histogram gradient boosting rather than implementing their Gradients

detector. No claim of equivalence to their method or their reported performance is made. Their preprint's version history starts in September 2024; the January 2025 version is the one cited here.

## 2.3 Confidence Is Not Confident Learning

Northcutt, Jiang, and Chuang's confident learning estimates the relationship between observed and latent labels using probabilistic thresholds and a class-conditional noise model [6]. It combines ranking with noise-distribution estimation. In contrast, this project's confidence score is simply one minus the out-of-fold probability of the observed label. Its cleaning budget is supplied from the experimental noise rate. It does not estimate the confident joint, implement the cleanlab algorithm, or establish the theoretical guarantees of confident learning. Confident learning is an important omitted comparator for future work.

## 2.4 Ground Truth and Statistical Evidence

De Vries and Thierens identify a difficulty with synthetic-noise experiments: adding known corruptions to a real dataset does not establish that its original labels were clean [7]. Their SYNLABEL framework constructs a ground-truth function and soft-label representations to make that distinction explicit. This project instead retains original public labels. Consequently, its detection metrics measure recovery of *injected changes*, not discovery of every genuinely incorrect original annotation.

Demšar emphasizes comparisons across datasets when evaluating learning algorithms [8]. This project adopts datasets as the aggregate inferential unit, not individual rows of its result matrix. It uses an exact paired sign-flip test rather than reproducing Demšar's recommended testing suite. The four datasets provide only limited inferential resolution; overlapping random splits and multiple conditions do not create hundreds of independent datasets.

## 2.5 Position of This Project

The distinguishing emphasis is traceability: each result can be connected to the split, corruption count, detector, cleaning action, seed, dependencies, and training revision that produced it. The comparison holds these elements fixed while changing the score or cleaning action. It is a focused empirical study, not a comprehensive competition against the literature. Its strongest supported conclusions concern relative behavior within this matrix and the failure of an intuitively appealing weighted score to beat a simpler baseline.

# Chapter 3

# Proposed Approach

## 3.1 Notation and Experimental Isolation

Let $D = \{(x_i, y_i)\}_{i=1}^{N}$ be a dataset with $C$ classes, represented by numeric feature vectors and reference labels. Stratified splitting produces training, validation, and test partitions in approximate proportions 60%, 20%, and 20%. The split seed is shared across all conditions belonging to a dataset–seed pair. The validation partition is reserved but not used for tuning or training in this study. Test labels are never corrupted or used to form cleaning scores.

Within the training partition, corruption produces observed labels $\widetilde{y}_i$ and a Boolean mask $z_i = \mathbf{1}[\widetilde{y}_i \neq y_i]$. The detectors observe $(x_i, \widetilde{y}_i)$, not $y_i$ or $z_i$. Reference training labels are used only for the clean baseline, corruption construction, and detection evaluation. This separation prevents a detector from exploiting the answer it is supposed to recover.

## 3.2 Controlled Label Corruption

### 3.2.1 Symmetric Noise

For a nominal corruption rate $\eta$ and $n$ training rows, sample $k = \text{round}(\eta n)$ distinct indices without replacement. Replace each selected label with a uniformly chosen member of the other $C - 1$ classes. Every selected label changes. In expectation, its transition probabilities are described by

$$T_{ab} = \Pr(\widetilde{y} = b \mid y = a) = \begin{cases} 1 - \eta, & a = b, \\ \eta/(C-1), & a \neq b. \end{cases} \tag{3.1}$$

The implementation fixes the total number of changes rather than sampling independent Bernoulli flips. Finite-sample transition frequencies therefore need not equal this idealized matrix exactly.

### 3.2.2 Cyclic Class-Dependent Noise

For sorted class labels $c_0, \ldots, c_{C-1}$, define $g(c_j) = c_{(j+1) \bmod C}$. Select $k_j = \text{round}(\eta n_j)$ rows from each original class and replace their labels using $g$. Class membership is computed before any labels are changed, so an example cannot be flipped twice as the class loop proceeds. The nominal transition matrix has diagonal mass $1 - \eta$ and off-diagonal mass $\eta$ at $g(c_j)$.

This model is class-dependent in the destination of a flip, not in a deliberately different error rate for each class. It is asymmetric for more than two classes, but binary cyclic and symmetric corruption both send a selected label to the only alternative class. Their binary implementations still differ in global versus per-class sample selection and rounding. The term "cyclic" is used to avoid implying a realistic annotator-confusion process. With per-class rounding, $\sum_j k_j$ can differ slightly from round($\eta n$); exact counts are saved in the result matrix.

## 3.3 Training-Only Out-of-Fold Probabilities

For each noisy dataset and base classifier, use shuffled stratified five-fold cross-validation within the training partition. Each training row receives a probability vector from a model fitted without that row:

$$p_i(c) = f^{(-v(i))}(x_i)_c, \tag{3.2}$$

where $v(i)$ denotes its fold. The seed for fold shuffling is the experiment seed. If a class has fewer than five rows, the implementation reduces the fold count to the smallest class count; fewer than two rows raises an error. The canonical data support the requested five folds.

Logistic regression's scaler is part of the fitted pipeline. Consequently, each fold estimates its own scaling parameters using only its fitting rows. The tree models use the numeric features without standardization. All four detectors share the same out-of-fold probability matrix, avoiding additional fitting variation or repeated cross-validation work. The same base model family supplies both the detector and the downstream classifier; this is not a cross-model transfer study.

## 3.4 Suspiciousness Scores

Define $\widehat{y}_i = \arg\max_c p_i(c)$, with the first sorted class chosen in an exact probability tie. Let $p_i^{(1)}$ and $p_i^{(2)}$ be the largest and second-largest predicted probabilities. Higher values of all four scores indicate greater suspicion:

$$s_i^{\text{conf}} = 1 - p_i(\widetilde{y}_i), \tag{3.3}$$

$$s_i^{\text{dis}} = \mathbf{1}[\widehat{y}_i \neq \widetilde{y}_i], \tag{3.4}$$

$$s_i^{\text{margin}} = 1 - (p_i^{(1)} - p_i^{(2)}), \tag{3.5}$$

$$s_i^{\text{comb}} = 0.5 s_i^{\text{conf}} + 0.3 s_i^{\text{dis}} + 0.2 s_i^{\text{margin}}. \tag{3.6}$$

All component scores lie in $[0, 1]$. The nonnegative weights sum to one and were fixed in the project design; they were not optimized using the test results. Confidence conditions directly on the observed label. Disagreement discards probability magnitude and creates many ties. Margin measures general uncertainty without asking whether the observed label is the uncertain class. That difference is central to interpreting the results.

### 3.4.1 Why the Binary Combination Adds Little Ranking Information

For two classes, let $s = 1 - p_i(\widetilde{y}_i)$. Away from a probability tie, substitution into Eq. (3.6) gives

$$s^{\text{comb}} = \begin{cases} 0.9s, & s < 0.5, \\ 0.1s + 0.7, & s > 0.5. \end{cases} \tag{3.7}$$

Both branches increase with $s$, and the second branch is above the first. Thus the combined score preserves the confidence ordering except for the handling of exact ties. At $s = 0.5$, the disagreement term can depend on which class wins the argmax tie. Matching binary performance is therefore expected and is not strong evidence that the combination contributes a new signal. For $C > 2$, the top-two margin is not determined by the observed-label probability, so the ranking can change. The multiclass results are the more informative component comparison.

## 3.5 Selection and Cleaning

For every detector, select the $k = \text{round}(\eta n)$ highest scores. Python's nearest-even rounding is used. Sorting is stable, ascending, and the last $k$ indices are selected. Exact ties at the cutoff therefore favor later rows in the current training-row order. That deterministic rule is especially consequential for the binary-valued disagreement score.

The use of $\eta$ is an *oracle selection budget*: the fraction of labels injected by the experiment is assumed known. The mask $z_i$ is not used for selection, but this budget assumption is more favorable than real deployment with an unknown noise level. For symmetric noise, the number of selected and corrupted rows is identical by construction. Detection precision and recall then coincide. Under cyclic corruption they may differ slightly because the number of injected errors is rounded separately within each class.

**Removal** discards selected examples and fits a fresh classifier on the remaining noisy-labeled training set. **Relabeling** keeps every feature row and replaces selected labels by $\widehat{y}_i$, then fits a fresh classifier. A selected row already predicted as its observed label remains unchanged after relabeling. Relabeling uses one model family's out-of-fold predictions, not an ensemble consensus or a second round of label recovery.

Two references complete the comparison: a fresh model trained on original training labels, and a fresh model trained on all noisy training labels. The original code identifier `clean_upper_bound` denotes the former; it is a reference condition, not a mathematical upper bound on test accuracy.

## 3.6 Experiment Algorithm and Cost

For each dataset, noise type, rate, model, and seed, the runner:

1. constructs the fixed training/validation/test split and training corruption;
2. fits and evaluates clean and noisy baseline models;
3. computes one shared out-of-fold probability matrix on noisy training data;
4. evaluates each detector against the injected-corruption mask;
5. removes or relabels its selected rows, retrains, and evaluates on the same test set;
6. writes the ten result rows and an atomic training-provenance checkpoint.

With five folds, each combination requires two baseline fits, five out-of-fold fits, and eight cleaned-model fits: 15 top-level estimator fits. Across 384 combinations this is 5,760 fits, excluding subsequent diagnostic retraining and reproducibility checks. A forest fit itself builds 200 trees. "Lightweight"

describes simple scores on conventional CPU estimators; it does not mean the full benchmark is cost-free or that runtime superiority was measured. The study does not provide a controlled timing comparison.

# Chapter 4

# Experimental Setup

## 4.1 Datasets and Preprocessing

Table 4.1 gives the exact benchmark dimensions and partition sizes. Breast Cancer Wisconsin (Diagnostic) contains image-derived nuclear measurements for binary diagnosis [9]. Wine contains chemical measurements associated with three wine classes [10]. Digits is scikit-learn's 1,797-row dataset of $8 \times 8$ digit images, flattened to 64 features [11]. It is a useful multiclass numeric benchmark, but not a conventional business table. It is a subset of Alpaydin and Kaynak's 5,620-row UCI optical-digit dataset [12]; this study re-splits the subset rather than reproducing the original writer-separated split. Dry Bean contains 16 measured shape-related features and seven classes [13].

**Table 4.1:** Dataset dimensions and exact split sizes.

| Dataset | $N$ | $p$ | $C$ | Train | Val. | Test |
|---|---|---|---|---|---|---|
| Breast Cancer | 569 | 30 | 2 | 341 | 114 | 114 |
| Wine | 178 | 13 | 3 | 106 | 36 | 36 |
| Digits | 1,797 | 64 | 10 | 1,078 | 359 | 360 |
| Dry Bean | 13,543 | 16 | 7 | 8,125 | 2,709 | 2,709 |

All features are numeric. $N$ is the post-deduplication sample count; $p$ is the number of features and $C$ the number of classes. Split rounding accounts for deviations from exact 60/20/20 proportions.

The first three datasets are loaded through scikit-learn. Dry Bean is downloaded from UCI and its ARFF representation is parsed with a structured reader. The original 13,611-row source is retained. Before splitting, 68 duplicate feature rows are removed, leaving 13,543 examples. Repeated feature vectors with conflicting labels are rejected rather than silently assigned a label. This prevents exact duplicate examples from appearing on both sides of the holdout boundary. The release audit checks disjoint row identities and feature signatures across all 16 dataset–seed splits.

No missing-value imputation, feature selection, resampling, or categorical encoding is needed for these data. Labels are encoded as integer class IDs. For Dry Bean the saved map is 0: BARBUNYA, 1: BOMBAY, 2: CALI, 3: DERMASON, 4: HOROZ, 5: SEKER, and 6: SIRA. No classification results are claimed for Adult Income or other unimplemented datasets.

## 4.2 Models and Fixed Parameters

All estimators use scikit-learn 1.6.1 [14]. The three model families span a linear classifier, bagged trees, and boosted trees. The random-forest inclusion is motivated by the established ensemble construction described by Breiman [15]. Hyperparameters are fixed rather than chosen separately for each noise condition.

**Table 4.2:** Estimator settings. Unlisted settings retain scikit-learn 1.6.1 defaults.

| Model | Settings |
|---|---|
| Logistic regression (LR) | StandardScaler; L2 penalty; $C = 1$; liblinear; one-vs-rest for multiclass; intercept enabled; maximum 2,000 iterations; tolerance $10^{-4}$; no class weighting. |
| Random forest (RF) | 200 trees; Gini criterion; bootstrap sampling; $\sqrt{p}$ features per split; unrestricted depth; minimum split size 2; minimum leaf size 1; no class weighting; all available jobs. |
| Histogram gradient boosting (HGB) | Log-loss; learning rate 0.1; maximum 100 iterations; 31 maximum leaf nodes; minimum leaf size 20; 255 maximum bins; L2 regularization 0; early stopping set to auto. |

The solver behavior and estimator defaults are documented in the versioned APIs [16–18]. For HGB, automatic early stopping requires more than 10,000 fitting rows. The largest training partition here has 8,125 rows, so this condition is never met. The reserved validation set is not passed to the estimator. Every fitted clone receives the experiment seed; cross-validation uses one job to avoid multiplying the forest's own parallel workers. Docker sets the OpenMP limit to four and the OpenBLAS and MKL thread limits to one.

## 4.3 Matrix and Paired Conditions

The experimental grid is

$$4 \text{ datasets} \times 3 \text{ models} \times 2 \text{ noise types} \times 4 \text{ rates} \times 4 \text{ seeds} = 384. \tag{4.1}$$

Rates are 10%, 20%, 30%, and 40%; seeds are 42, 123, 999, and 2026. Each combination produces two reference rows plus two cleaning actions for each of four detectors: $384(2 + 2 \cdot 4) = 3,840$ rows. Clean references recur across noise conditions for convenience. Those repeated values are not independent experiments. Likewise, removal and relabeling share a detector's mask; their detection scores are counted once per combination and detector.

Identical model–dataset–noise–rate–seed keys are used when comparing a cleaned model with its noisy reference. Comparisons between the combined score and its component scorers also hold the cleaning action fixed. The benchmark is balanced, so simple grand means give equal weight to each dataset, model, noise type, rate, and seed. Large datasets do not dominate the aggregate solely because they contain more rows.

## 4.4 Evaluation Metrics

### 4.4.1 Downstream Classification

Accuracy is the proportion of correct test predictions. Macro F1 computes $F_{1,c} = 2P_cR_c/(P_c+R_c)$ for each class and averages over the classes, so minority classes do not disappear into a sample-weighted mean. A zero denominator is assigned zero. AUROC uses the probability of the positive class for binary classification. For multiclass classification, it averages one-vs-rest class AUROCs with equal class weight. The same untouched test partition is used for every condition in a dataset–seed pair.

### 4.4.2 Injected-Error Detection

Let $S$ be the selected set and $Z$ the known set of injected corruptions. Detection precision is $|S \cap Z|/|S|$, recall is $|S \cap Z|/|Z|$, and F1 is their harmonic mean. These are not test-set classification metrics and do not establish which original annotations were incorrect. Undefined ratios are assigned zero by the implementation. The canonical positive-noise grid contains nonempty selection and corruption sets.

### 4.4.3 Diagnostic Curves

Saved-model diagnostics supplement the matrix with confusion matrices, per-class classification reports, ROC curves, and precision–recall (PR) curves. Multiclass diagnostic curves flatten one-hot targets and predicted probabilities to produce *micro-averaged* curves. Their ROC area is therefore not the main table's macro one-vs-rest AUROC. PR area is computed by trapezoidal integration, not by average precision. These choices are named explicitly wherever the curves or areas appear.

## 4.5 Aggregation and Inference

For method $a$, dataset $d$, setting $s$, and seed $r$, define its paired gain over noisy training as

$$\delta_{dsr}^{(a)} = M_{dsr}^{(a)} - M_{dsr}^{(\text{noisy})}. \tag{4.2}$$

The study first averages the four seeds within a setting, then the 24 settings within a dataset, and finally the four dataset means:

$$g_d^{(a)} = \frac{1}{24}\sum_{s=1}^{24}\frac{1}{4}\sum_{r=1}^{4}\delta_{dsr}^{(a)}, \qquad \overline{g}^{(a)} = \frac{1}{4}\sum_{d=1}^{4} g_d^{(a)}. \tag{4.3}$$

The dataset-level unit follows the comparative-evaluation motivation in Demšar [8]; it does not make the four convenience datasets a random sample of all tabular tasks.

The exact paired sign-flip test enumerates $2^4 = 16$ sign assignments to the four gains and uses the mean as its statistic. It is implemented through SciPy's one-sample paired-permutation interface [19]. Its interpretation requires exchangeability of paired method assignments, or a corresponding sign-symmetry null for dataset effects. This is an exploratory assumption, not a property guaranteed by the dataset selection. The smallest attainable two-sided $p$-value with four nonzero gains is $2/16 = 0.125$. Statistical significance at 0.05 is therefore unattainable in this design, even before multiplicity adjustment.

Holm's sequential correction is applied within each scope and metric [20]: eight cleaning-versus-noisy comparisons form one family, and six combined-versus-component comparisons form another. These are not a single adjustment across every metric and family. Exploratory 95% percentile bootstrap intervals use 5,000 resamples of the four dataset means with random seed 2026. They have limited reliability with four datasets and must not override the exact-test resolution. Per-dataset intervals and $p$-values are left undefined. Win/tie/loss percentages use 96 seed-averaged settings, a $10^{-12}$ tie tolerance, and remain descriptive.

## 4.6 Execution and Verification

The canonical matrix was trained under Docker on Linux ARM64 with CPython 3.11.17. The pinned analysis stack includes NumPy 1.26.4, pandas 2.3.3, scikit-learn 1.6.1, SciPy 1.13.1, and Matplotlib 3.9.4. Native macOS testing also exercised the Python 3.9 environment. The original training revision is distinct from the later audit and reporting revisions.

At the frozen code/results revision, 133 automated tests passed in both environments, with 88% branch-aware core coverage on the host. The release audit checked 25 result-contract rules, split isolation, corruption counts, table recomputation, saved-model metric reproduction, image presence, and 181 artifact hashes. A fresh extraction and container build passed the same audit offline. Additionally, all 24 dataset–model–noise combinations at 40% noise and seed 999 were rerun; their 240 result rows matched the saved matrix to numerical tolerance $10^{-12}$. These checks reduce implementation risk; they do not prove the scientific assumptions or guarantee bug-free code.

# Chapter 5

# Results and Analysis

## 5.1 Aggregate Classification Performance

Table 5.1 summarizes the complete balanced matrix. Mean accuracy falls from 0.9586 with original training labels to 0.8535 with noisy labels, a loss of approximately 10.52 percentage points. Confidence removal recovers 4.27 points, while combined removal recovers 3.68 points. Neither restores the clean reference on average. The corresponding macro-F1 and AUROC values also favor confidence removal among the evaluated cleaning methods.

**Table 5.1:** Overall test performance across the balanced matrix.

| Method | Accuracy | Macro F1 | AUROC | Δ acc. (pp) |
|---|---|---|---|---|
| Clean reference | 0.9586 | 0.9614 | 0.9956 | +10.52 |
| Noisy baseline | 0.8535 | 0.8520 | 0.9519 | +0.00 |
| Confidence / Remove | 0.8961 | 0.8968 | 0.9723 | +4.27 |
| Confidence / Relabel | 0.8904 | 0.8909 | 0.9699 | +3.69 |
| Combined / Remove | 0.8903 | 0.8910 | 0.9708 | +3.68 |
| Combined / Relabel | 0.8836 | 0.8839 | 0.9676 | +3.01 |
| Disagreement / Remove | 0.8834 | 0.8834 | 0.9662 | +3.00 |
| Disagreement / Relabel | 0.8720 | 0.8718 | 0.9618 | +1.86 |
| Margin / Remove | 0.7939 | 0.7892 | 0.9235 | -5.96 |
| Margin / Relabel | 0.8279 | 0.8254 | 0.9389 | -2.56 |

Each method has 384 rows. Means weight datasets, models, noise types, rates, and seeds equally. "pp" denotes percentage points. Repeated clean values are a reference, not independent measurements.

Relabeling is beneficial on average for confidence, combined, and disagreement scoring, but their removal variants achieve higher aggregate accuracy. Margin-only removal is the worst intervention: accuracy is 0.7939, below the noisy baseline by 5.96 points. Margin relabeling also reduces accuracy, by 2.56 points. Thus applying a nominally plausible uncertainty score is not sufficient to improve robustness; the relationship between the score and the *observed label* matters.

## 5.2 RQ1: Sensitivity to Noise

Table 5.2 separates the accuracy loss by dataset, noise mechanism, and rate. The clean and noisy models are paired before aggregation. On Dry Bean, the mean loss under symmetric 40% corruption

is 2.92 points, compared with 20.75 points under cyclic corruption. Digits shows the same contrast: 8.82 versus 29.26 points. This suggests that a concentrated wrong-class transition is more damaging than dispersing errors among many alternatives for these multiclass tasks. It does not establish a universal ordering of noise mechanisms, because only one cyclic map is evaluated.

**Table 5.2:** Accuracy loss from clean to noisy training, in percentage points.

| Dataset | Noise | 10% | 20% | 30% | 40% |
|---|---|---|---|---|---|
| Breast Cancer | Symmetric | 2.78 | 5.70 | 14.62 | 25.66 |
| Breast Cancer | Cyclic | 1.32 | 4.46 | 11.26 | 29.17 |
| Wine | Symmetric | 6.48 | 9.72 | 15.05 | 29.40 |
| Wine | Cyclic | 4.17 | 11.34 | 20.14 | 33.80 |
| Digits | Symmetric | 1.41 | 3.38 | 6.71 | 8.82 |
| Digits | Cyclic | 2.52 | 5.86 | 14.75 | 29.26 |
| Dry Bean | Symmetric | 0.36 | 0.86 | 1.65 | 2.92 |
| Dry Bean | Cyclic | 0.77 | 2.93 | 8.52 | 20.75 |

Each cell averages the paired loss across three models and four seeds. Rates are nominal training-label corruption fractions.

For Breast Cancer, both mechanisms are binary flips, and the differences reflect sampled rows and per-class versus global corruption rather than different destination labels. Wine is especially sensitive at high noise rates: its small training and test partitions limit precision. One changed test prediction on Wine corresponds to 1/36, or about 2.78 accuracy points in a single run. Small differences should not be interpreted as strong rankings of methods.

Figures 5.1 and 5.2 show selected conditions across rates. They display averages, not uncertainty bands. The complete release contains per-model plots with seed-standard-deviation shading; those shaded regions are not confidence intervals over independent datasets.

### 5.2.1 Variation across Model Families

Table 5.3 reports dataset–model means across rates, mechanisms, and seeds. RF has the strongest mean noisy-label accuracy on Wine and Digits; LR has the strongest on Dry Bean. HGB has lower mean noisy accuracy than both other models on each of the four datasets. This is evidence about the fixed settings used here, not about optimally tuned boosting in general. For Wine, HGB rises from 0.7439 to 0.8429 after confidence removal but remains below the cleaned RF value of 0.9418.

Removal is not always preferable to relabeling. On Digits with RF, confidence relabeling reaches 0.9370 versus 0.9330 for removal. On Dry Bean with HGB, the corresponding means are nearly tied. A global average therefore does not justify selecting one action blindly for every model–dataset pair.

## 5.3 RQ2: Detection of Injected Errors

Table 5.4 counts each detector once per experimental combination, not once for both removal and relabeling. Confidence has the highest average detection F1, 0.8072, followed by combined, 0.7504, disagreement, 0.6827, and margin, 0.2425. Because the selected fraction closely matches the injected fraction, precision and recall are nearly equal; this is partly a design property rather than independent evidence from two unrelated criteria.

Margin-only selection behaves differently from confidence selection. A confidently wrong observed

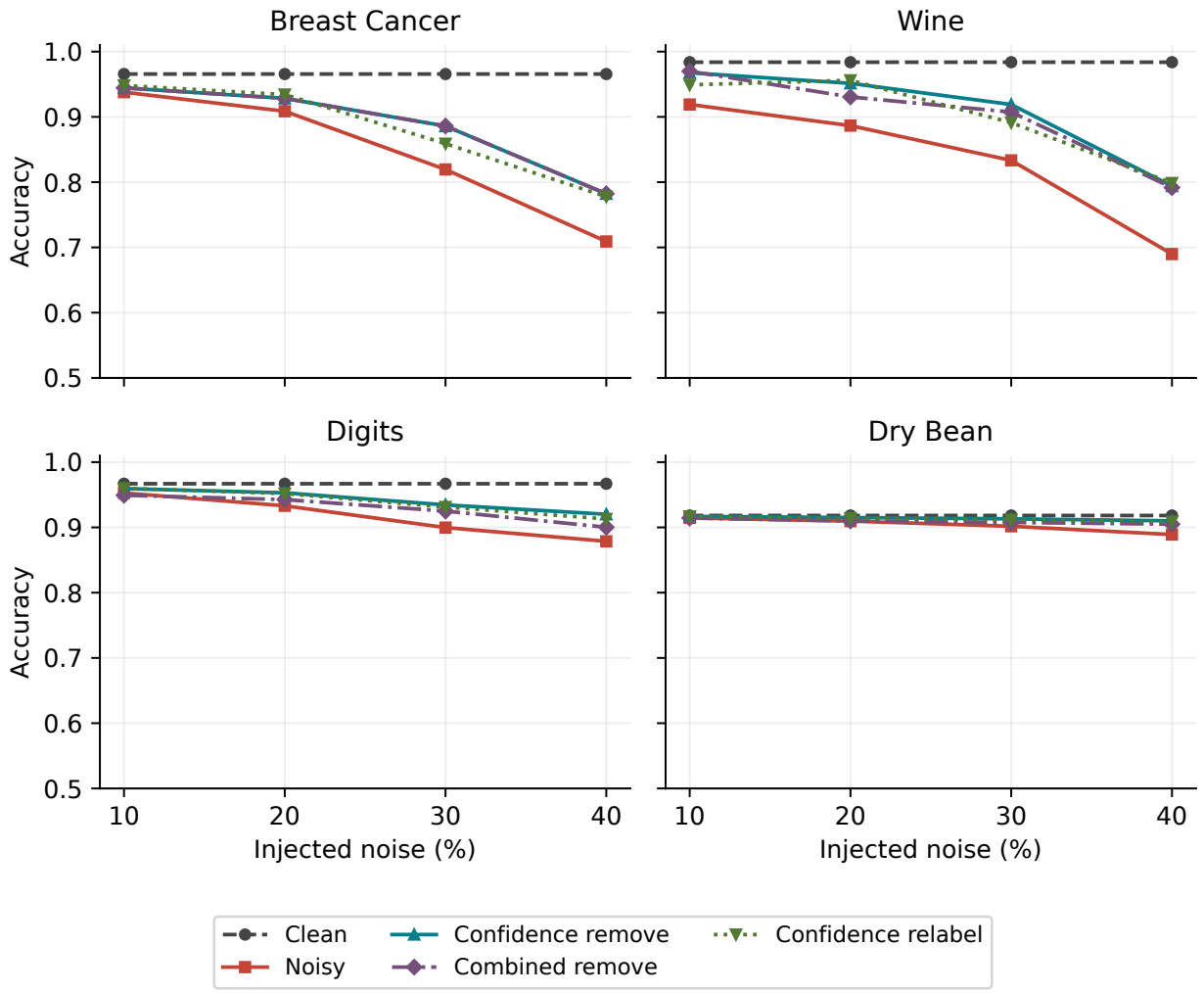



**Figure 5.1:** Accuracy under symmetric corruption. Each point averages three models and four seeds within a dataset. Selected methods are shown for legibility; all methods appear in Table 5.1. Lines connect the four evaluated rates and do not represent fitted continuous trends.

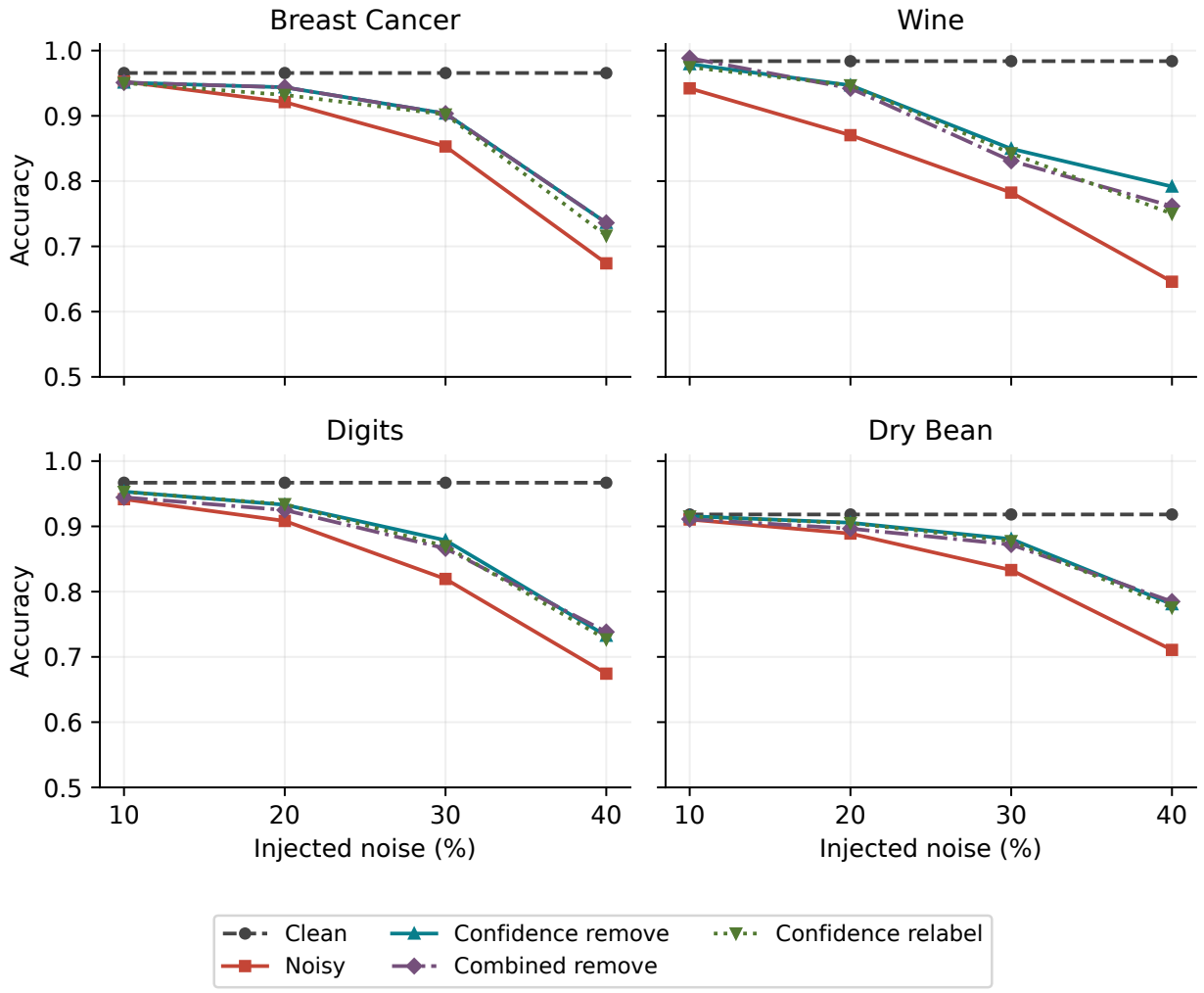



**Figure 5.2:** Accuracy under cyclic class-dependent corruption, aggregated as in Fig. 5.1. High-rate cyclic corruption is particularly damaging on the multiclass tasks.

**Table 5.3:** Mean test accuracy by dataset and classifier.

| Dataset | Model | Clean | Noisy | Conf. rem. | Conf. rel. | Comb. rem. |
|---|---|---|---|---|---|---|
| Breast Cancer | LR | 0.9803 | 0.8660 | 0.8961 | 0.8871 | 0.8961 |
| Breast Cancer | RF | 0.9539 | 0.8706 | 0.8972 | 0.8958 | 0.8972 |
| Breast Cancer | HGB | 0.9627 | 0.8043 | 0.8599 | 0.8495 | 0.8599 |
| Wine | LR | 0.9861 | 0.8464 | 0.9149 | 0.9010 | 0.9062 |
| Wine | RF | 0.9931 | 0.8733 | 0.9418 | 0.9340 | 0.9288 |
| Wine | HGB | 0.9722 | 0.7439 | 0.8429 | 0.8307 | 0.8359 |
| Digits | LR | 0.9639 | 0.8477 | 0.8872 | 0.8774 | 0.8718 |
| Digits | RF | 0.9722 | 0.9198 | 0.9330 | 0.9370 | 0.9319 |
| Digits | HGB | 0.9646 | 0.8605 | 0.9041 | 0.9003 | 0.8930 |
| Dry Bean | LR | 0.9140 | 0.8806 | 0.8899 | 0.8861 | 0.8773 |
| Dry Bean | RF | 0.9206 | 0.8703 | 0.8957 | 0.8946 | 0.8953 |
| Dry Bean | HGB | 0.9201 | 0.8584 | 0.8909 | 0.8910 | 0.8904 |

LR: logistic regression; RF: random forest; HGB: histogram gradient boosting. Each cell averages two noise types, four rates, and four seeds. "Conf." and "Comb." denote confidence and combined scoring; rem./rel. denote removal/relabeling.

**Table 5.4:** Injected-label-error detection, averaged over 384 combinations per detector.

| Detector | Precision | Recall | F1 |
|---|---|---|---|
| Confidence | 0.8077 | 0.8067 | 0.8072 |
| Combined | 0.7509 | 0.7500 | 0.7504 |
| Disagreement | 0.6832 | 0.6822 | 0.6827 |
| Margin | 0.2427 | 0.2422 | 0.2425 |

Detection is evaluated on training rows. Removal and relabeling share each detector's mask and are not double-counted.

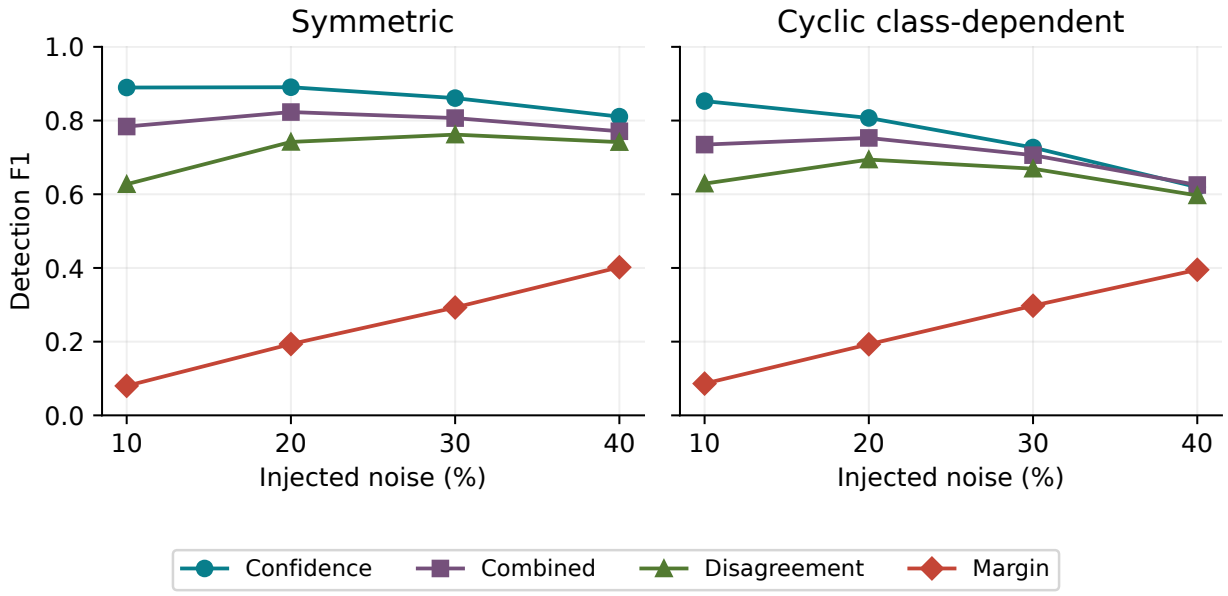



**Figure 5.3:** Detection F1 by nominal rate and corruption mechanism. Each point averages the four datasets, three models, and four seeds. No test labels are used to compute a suspiciousness score.

label can receive a low margin score when the model strongly predicts another class. Conversely, a correctly labeled but ambiguous example can receive a high margin score. The poor detection and downstream outcomes are consistent with this mismatch between uncertainty and label error. This is a mechanism-based interpretation of the scores, not a causal experiment isolating boundary examples.

## 5.4 RQ3: Effects of Cleaning and Relabeling

Figure 5.4 shows the paired accuracy gain within each dataset. Confidence removal improves all four dataset-level averages, from 2.24 points on Dry Bean to 7.87 points on Wine. Margin removal reduces all four averages, from 4.14 points on Dry Bean to 7.12 points on Digits. This pattern is not merely the result of pooling the largest dataset with smaller ones: datasets receive equal weight and their separate effects are visible.

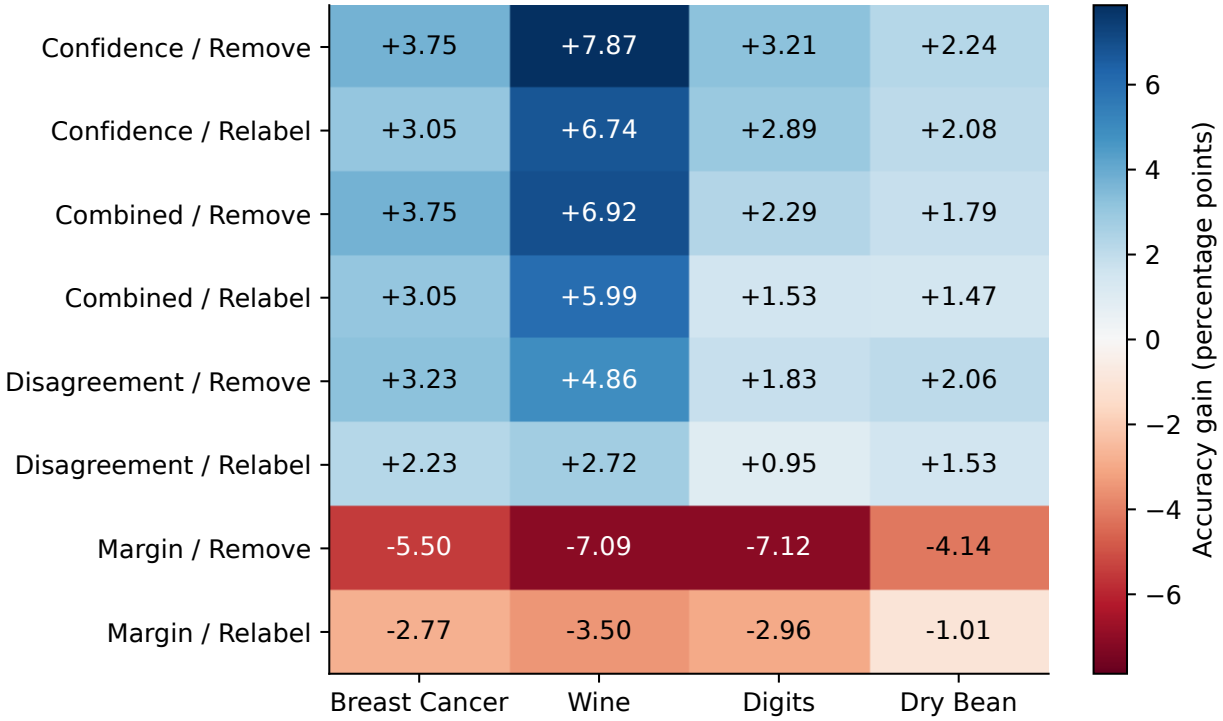



**Figure 5.4:** Paired accuracy gains over noisy training, in percentage points. Every cell averages 24 settings and four seeds for one dataset. Numeric annotations preserve interpretation without color.

Table 5.5 reports both effect size and the limits of inference. Confidence removal improves 94.8% of the 96 seed-averaged settings. This is not the same as its proportion of individually improved seed runs: aggregation reduces some seed variability before a setting is counted as a win. The confidence-removal exploratory interval is entirely positive, yet the exact two-sided $p$-value is 0.125 and its Holm-adjusted value is 1.000. There is no contradiction: the bootstrap percentile interval and discrete exact test answer the uncertainty question differently with very sparse dataset-level evidence. The interval must not be used to claim significance.

**Table 5.5:** Paired accuracy effects and exploratory dataset-level inference.

| Method | Gain (pp) | Exploratory 95% CI | Win % | $p$ | $p_H$ |
|---|---|---|---|---|---|
| Confidence / Remove | +4.27 | [+2.62, +6.70] | 94.8 | 0.125 | 1.000 |
| Confidence / Relabel | +3.69 | [+2.32, +5.78] | 92.7 | 0.125 | 1.000 |
| Combined / Remove | +3.68 | [+2.04, +5.76] | 79.2 | 0.125 | 1.000 |
| Combined / Relabel | +3.01 | [+1.50, +4.87] | 76.0 | 0.125 | 1.000 |
| Disagreement / Remove | +3.00 | [+1.95, +4.16] | 78.1 | 0.125 | 1.000 |
| Disagreement / Relabel | +1.86 | [+1.24, +2.47] | 67.7 | 0.125 | 1.000 |
| Margin / Relabel | -2.56 | [-3.32, -1.50] | 10.4 | 0.125 | 1.000 |
| Margin / Remove | -5.96 | [-7.10, -4.82] | 2.1 | 0.125 | 1.000 |

Gain and interval endpoints are percentage points. Intervals resample four dataset means. Win rates use 96 seed-averaged settings. $p_H$ is Holm-adjusted within the eight accuracy comparisons. None is significant at 0.05.

### 5.4.1 Does the Combined Score Help?

The direct comparison in Table 5.6 gives combined removal an average deficit of 0.58 accuracy points relative to confidence removal; combined relabeling has a deficit of 0.68 points. Combined scoring exceeds disagreement and margin on the grand mean, but does not beat the best simple component. All reported adjusted comparisons remain nonsignificant.

**Table 5.6:** Combined scoring minus each component detector under the same action.

| Reference detector | Action | Combined gain (pp) | $p$ | $p_H$ |
|---|---|---|---|---|
| Confidence | Remove | -0.58 | 0.250 | 1.000 |
| Confidence | Relabel | -0.68 | 0.250 | 1.000 |
| Disagreement | Relabel | +1.15 | 0.250 | 1.000 |
| Disagreement | Remove | +0.69 | 0.250 | 1.000 |
| Margin | Remove | +9.65 | 0.125 | 0.750 |
| Margin | Relabel | +5.57 | 0.125 | 0.750 |

Positive gain favors combined. Each comparison uses matched dataset–model–noise–rate–seed observations; inference uses four dataset means. Holm adjustment covers the six accuracy comparisons.

Breast Cancer's combined and confidence downstream results coincide in the aggregate, as the binary ranking analysis predicts. For each of the three multiclass datasets, confidence has a larger mean gain for both actions. The combination can shift highly uncertain examples upward without providing additional evidence that their observed labels are wrong. A different weight choice might behave differently, but choosing it after inspecting these test results would require a new, independently evaluated selection procedure. The present negative result should not be obscured by retrospective tuning.

### 5.4.2 A Concrete Failure despite Good Detection

The saved-model example uses Dry Bean, RF, symmetric 20% noise, confidence scoring, and seed 42. Table 5.7 makes the failure case explicit. Detection F1 is 0.9151, but removal and relabeling both reduce test accuracy relative to the noisy baseline. They improve macro AUROC in this setting, showing that probability ranking and hard-label accuracy can move in different directions. The small differences do not support a separate statistical conclusion from one seed.

The noisy classifier's confusion matrix in Fig. 5.5 identifies where residual errors occur. Class

**Table 5.7:** Illustrative saved-model setting: Dry Bean/RF/symmetric 20%/seed 42.

| Condition | Accuracy | Macro F1 | Macro AUROC | Micro PR AUC |
|---|---|---|---|---|
| Clean reference | 0.9203 | 0.9313 | 0.9915 | 0.9734 |
| Noisy baseline | 0.9169 | 0.9280 | 0.9874 | 0.9558 |
| Confidence / Remove | 0.9136 | 0.9261 | 0.9899 | 0.9675 |
| Confidence / Relabel | 0.9133 | 0.9268 | 0.9897 | 0.9661 |

Confidence is used for both cleaning actions. PR AUC is the trapezoidal area under the micro-averaged precision–recall curve, not average precision.

IDs are defined in Chapter 4. This example is a diagnostic illustration, not an unbiased choice of the best-performing setting. The full matrix, rather than this single case, supports the aggregate comparison. ROC and PR curves for the same saved classifier are provided in Appendix A.

## 5.5 Threats to Validity

**Internal validity.** Shared splits, training-only out-of-fold scoring, pipeline-local scaling, exact-count corruption, and duplicate removal reduce leakage risk. Saved checksums and re-execution provide evidence that the tables correspond to the code. Nevertheless, testing cannot exclude every implementation error. The same integer seed drives multiple random processes; results use deterministic paired conditions rather than independently randomized substreams. A future design could study sensitivity to separate split, corruption, and estimator seeds.

**Construct validity.** The injected mask is an operational definition of noise. It does not establish that the original labels are correct. The cyclic transition is arbitrary in class-ID space, and the known-rate budget is unavailable in many applications. Margin measures uncertainty, while confidence combines possible label error with model misspecification. The reserved validation data are unused; this avoids tuning leakage but does not evaluate practical threshold selection.

**External validity.** Four numeric datasets, only one binary task, and no real audited annotation errors limit generalization. Digits is image-derived and the other datasets are relatively small. There are no mixed categorical tables, deployment shifts, feature-noise experiments, or calibrated estimates of label-error prevalence. No clinical or other high-stakes use is supported by these experiments. Random stratified partitions do not establish generalization to unseen patients, writers, or collection sites; subject-group independence is not evaluated by this benchmark.

**Comparative validity.** Only fixed estimator parameters and four simple scores are compared. Confident learning, robust losses, stronger tabular learners, instance-dependent corruption, and adaptive budgets remain unmeasured. No runtime benchmark establishes that the proposed pipeline is faster than alternatives. The combined score's modest engineering complexity is not evidence of a new algorithmic contribution.

**Statistical validity.** The 3,840 rows share datasets, splits, corruptions, and repeated references. They cannot be treated as independent replicates. Four dataset means are insufficient for precise population-level inference; the convenience sample also weakens generalization beyond the observed tasks. Retrospective best-by-setting tables are descriptive only and are not an evaluated model-selection policy.

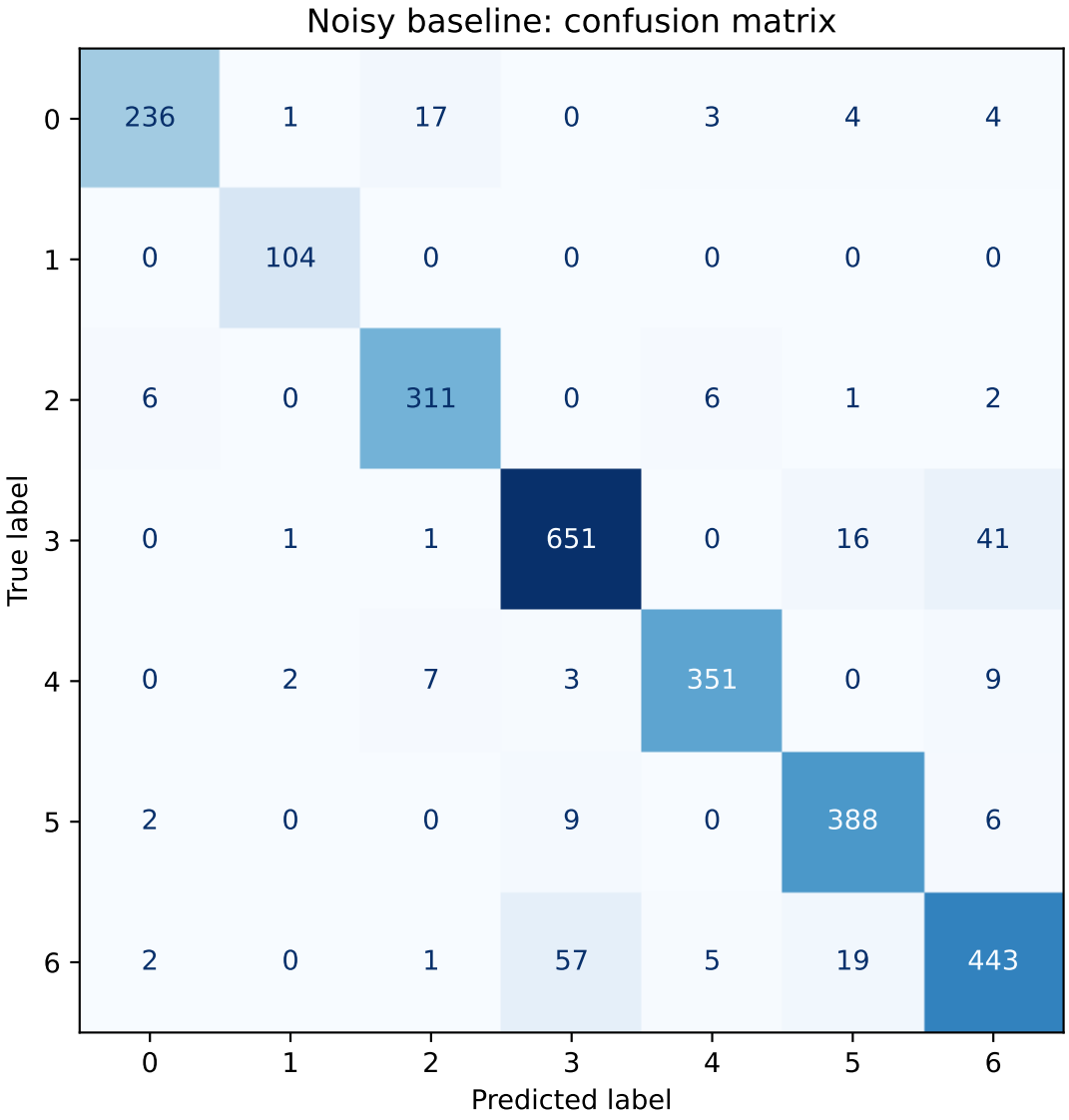



**Figure 5.5:** Test confusion matrix for the illustrative noisy RF classifier. Rows are reference classes and columns predicted classes. The test set has 2,709 examples and was not corrupted.

# Chapter 6

# Conclusion and Future Work

## 6.1 Answers to the Research Questions

**RQ1:** Training-label corruption reduces aggregate performance and its effect depends strongly on dataset, model, and transition structure. The mean clean-reference accuracy is 0.9586, compared with 0.8535 for noisy training. High-rate cyclic corruption is particularly damaging on Digits and Dry Bean. Fixed-default HGB is more sensitive than the other evaluated families on the dataset-level means, but this is not a conclusion about optimally tuned boosting.

**RQ2:** Out-of-fold confidence is the most effective of the four evaluated detectors on average, with detection F1 of 0.8072. Combined scoring follows at 0.7504. General uncertainty, measured by the top-two probability margin, is a poor proxy for injected label errors in this design. The shared out-of-fold matrix and common selection budget make the component comparison directly interpretable.

**RQ3:** Confidence-based removal improves average test accuracy by 4.27 percentage points and combined removal by 3.68 points. Neither closes the full gap to clean training. Relabeling helps on average for three detectors, but is not uniformly better than removal. High detection quality does not guarantee a downstream accuracy gain, as the saved Dry Bean example demonstrates. Margin-only cleaning is harmful on all four dataset-level averages.

## 6.2 Contribution Supported by the Evidence

The project delivers a reproducible benchmark and a transparent evaluation of a simple combined-score hypothesis. The hypothesis does not succeed against confidence alone on the aggregate comparison. The binary analysis explains why the combination often cannot change confidence's ranking; the multiclass experiments show that changing the ranking does not necessarily improve it. These results favor starting with a simple confidence baseline before adding uncertainty terms, within the controlled setting tested here.

The release includes an auditable matrix, generated tables and figures, illustrative saved models, tests, pinned dependencies, and an immutable Git-LFS archive. This supports verification and extension of the experiment, not a claim of universal method superiority. With four datasets, no exact two-sided comparison can reach significance at 0.05. Conclusions remain descriptive and exploratory.

## 6.3 Priorities for Future Work

The first priority is to remove the oracle-budget assumption. A future study should choose thresholds or removal fractions using only training information or a properly allocated validation set, then evaluate the complete selection procedure on untouched test data. This would test whether the current gains survive uncertainty about the actual noise prevalence.

Second, the comparison should add established label-quality methods such as confident learning, alternative cyclic maps, and feature-dependent corruptions. Real annotation-error datasets would distinguish injected-error recovery from genuine data-quality improvement. Additional mixed-type tabular datasets would broaden the empirical population and improve the resolution of dataset-level inference.

Third, model and score choices should be evaluated under a prespecified tuning protocol. Probability calibration, class-aware selection budgets, and confidence-gated relabeling are reasonable hypotheses, but should not be selected from the current test outcomes and presented as already validated. Randomized tie handling and separately seeded random processes are additional sensitivity checks. Finally, controlled runtime and memory measurements would turn the qualitative term "lightweight" into a measured cost comparison.

# References


[1] B. Frénay and M. Verleysen, "Classification in the presence of label noise: A survey," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 25, no. 5, pp. 845–869, 2014. [Online]. Available: https://doi.org/10.1109/TNNLS.2013.2292894

[2] L. Nazaretyan, U. Leser, and M. Kircher, "Benchmarking machine learning methods for the identification of mislabeled data," *Artificial Intelligence Review*, vol. 58, Jul. 2025, art. no. 310. [Online]. Available: https://doi.org/10.1007/s10462-025-11293-9

[3] M. Li and C. Zhu, "Noisy label processing for classification: A survey," arXiv:2404.04159, Apr. 2024, preprint. [Online]. Available: https://arxiv.org/abs/2404.04159

[4] J. Yang, H. Triendl, A. A. S. Soltan, M. Prakash, and D. A. Clifton, "Addressing label noise for electronic health records: insights from computer vision for tabular data," *BMC Medical Informatics and Decision Making*, vol. 24, Jun. 2024, art. no. 183. [Online]. Available: https://doi.org/10.1186/s12911-024-02581-5

[5] A. Eisenbürger, D. Otten, A. Hudde, and F. Hopfgartner, "Training gradient boosted decision trees on tabular data containing label noise for classification tasks," arXiv:2409.08647v2, Jan. 2025, preprint; version 2, Jan. 6, 2025. [Online]. Available: https://arxiv.org/abs/2409.08647v2

[6] C. Northcutt, L. Jiang, and I. Chuang, "Confident learning: Estimating uncertainty in dataset labels," *Journal of Artificial Intelligence Research*, vol. 70, pp. 1373–1411, 2021. [Online]. Available: https://doi.org/10.1613/jair.1.12125

[7] S. de Vries and D. Thierens, "Generating the ground truth: Synthetic data for soft label and label noise research," *International Journal of Data Science and Analytics*, vol. 20, pp. 5603–5615, 2025. [Online]. Available: https://doi.org/10.1007/s41060-025-00786-z

[8] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *Journal of Machine Learning Research*, vol. 7, pp. 1–30, 2006. [Online]. Available: https://jmlr.org/papers/v7/demsar06a.html

[9] UCI Machine Learning Repository, "Breast Cancer Wisconsin (Diagnostic)," Dataset, 1993, accessed: Oct. 3, 2026. [Online]. Available: https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic

[10] S. Aeberhard and M. Forina, "Wine," UCI Machine Learning Repository, dataset, 1992, accessed: Oct. 3, 2026. [Online]. Available: https://archive.ics.uci.edu/dataset/109/wine

[11] scikit-learn Developers, "load_digits," scikit-learn 1.6.1 documentation, accessed: Oct. 3, 2026. [Online]. Available: https://scikit-learn.org/1.6/modules/generated/sklearn.datasets.load_digits.html

[12] E. Alpaydin and C. Kaynak, "Optical Recognition of Handwritten Digits," UCI Machine Learning Repository, dataset, 1998, accessed: Oct. 3, 2026. [Online]. Available: https://archive.ics.uci.edu/dataset/80/optical+recognition+of+handwritten+digits

[13] UCI Machine Learning Repository, "Dry Bean," Dataset, 2020, dataset associated with M. Koklu and I. A. Ozkan. Accessed: Oct. 3, 2026. [Online]. Available: https://archive.ics.uci.edu/dataset/602/dry+bean+dataset

[14] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011. [Online]. Available: https://jmlr.org/papers/v12/pedregosa11a.html

[15] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001. [Online]. Available: https://doi.org/10.1023/A:1010933404324

[16] scikit-learn Developers, "LogisticRegression," scikit-learn 1.6.1 documentation, accessed: Oct. 3, 2026. [Online]. Available: https://scikit-learn.org/1.6/modules/generated/sklearn.linear_model.LogisticRegression.html

[17] ——, "RandomForestClassifier," scikit-learn 1.6.1 documentation, accessed: Oct. 3, 2026. [Online]. Available: https://scikit-learn.org/1.6/modules/generated/sklearn.ensemble.RandomForestClassifier.html

[18] ——, "HistGradientBoostingClassifier," scikit-learn 1.6.1 documentation, accessed: Oct. 3, 2026. [Online]. Available: https://scikit-learn.org/1.6/modules/generated/sklearn.ensemble.HistGradientBoostingClassifier.html

[19] SciPy Developers, "scipy.stats.permutation_test," SciPy 1.13.1 documentation, accessed: Oct. 3, 2026. [Online]. Available: https://docs.scipy.org/doc/scipy-1.13.1/reference/generated/scipy.stats.permutation_test.html

[20] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979. [Online]. Available: https://www.jstor.org/stable/4615733

# Appendix A

# Reproducibility and Supplementary Diagnostics

## A.1 Immutable Evidence

The report is based on the code/results release at revision:

`50f8995a014f4d68714cba7a4ea81e911c859374`

The original training revision is:

`2d0dfbf34480bba5e602f165fd12dc0c3ef75ef8`

Later report commits do not rewrite the historical training record.

The main result CSV contains 3,840 rows and has SHA-256

`2576804e3caadd3025ff95a9835729ced219fd38dce4ca30ee4a399727a61150.`

The archive `releases/code-results-50f8995.tar.gz` has SHA-256

`73b15ce8a707f671f7d7102d8ac776d504a9786afda5d933f0a72e7363861e58.`

It contains 264 source and artifact files, including the 181 items in the frozen audit inventory. The source repository is `https://github.com/Ovais09/csi6900-noisy-labels`. Git LFS stores the large archive separately from the ordinary Git pointer. It must be fetched before extracting the release. The archive must not be unpacked over an active working tree, and untrusted serialized models must not be loaded.

## A.2 Artifact Map

| Artifact | Purpose |
|---|---|
| `configs/experiment_config.yaml` | Exact experimental grid. |

| Artifact | Purpose |
|---|---|
| `results/tables/main_experiment_results.csv` | Per-combination and per-condition metrics and counts. |
| `results/tables/summary_results.csv` | 960 rows aggregating the four seeds within settings. |
| `results/tables/analysis/` | Eight analysis tables, including paired cleaning and direct detector comparisons. |
| `results/models/` | Four RF models for one Dry Bean setting, not all models fitted by the benchmark. |
| `results/figures/` | 60 main and 13 diagnostic figures, each in PNG and PDF. |
| `results/logs/run_manifest.json` | Config, data, environment, result, and original training provenance. |
| `results/logs/release_manifest.json` | Artifact sizes and SHA-256 hashes. |
| `results/logs/final_audit.json` | Passing audit of that frozen inventory. |
| `report/generated/evidence.json` | Checksummed inputs used to generate report tables and figures. |

The report generator recomputes analysis tables from the exact CSV and checks them against the frozen versions before producing LaTeX tables. Numeric summary tables are not manually transcribed. Report-specific plots average the same matrix; selected diagnostic PDFs are copied unchanged after hash verification. Their source values and revision are recorded in the evidence file. The IEEEtran BibTeX style produces the numbered bibliography.

## A.3 Build and Reproduction Commands

The complete extraction and audit commands are in `releases/README.md`. The following commands assume the Git-LFS archive has been verified and restored to an empty `releases/restored/` directory:

```
python scripts/make_report_assets.py --release-root releases/restored
docker build -f report/Dockerfile -t csi6900-report .
docker run --rm --network none -v "$PWD:/work" csi6900-report
```

The last command runs pdfLaTeX and BibTeX through latexmk and writes `report/build/main.pdf`. Building from the already committed report assets does not require model training or downloading datasets. A full benchmark rerun uses the archived configuration and pipeline in its own checkout; it should not overwrite the frozen evidence used by this report.

## A.4 Preparation and Review

AI assistance was used during software implementation, testing, analysis support, and manuscript preparation. The numerical evidence is tied to the saved experiment artifacts, not generated as illustrative data. Source verification and software checks do not replace the author's responsibility to understand the methods, review the conclusions, and comply with the supervisor's and institution's requirements for assistance and disclosure.

## A.5 Supplementary Macro F1 and Curves

Figure A.1 provides the macro-F1 counterpart of the cyclic accuracy plot. Figures A.2 and A.3 concern only the saved noisy RF classifier from the Dry Bean diagnostic setting. Their smooth ranking summaries should not be interpreted as proof that cleaning improves the hard-label predictions in that setting.

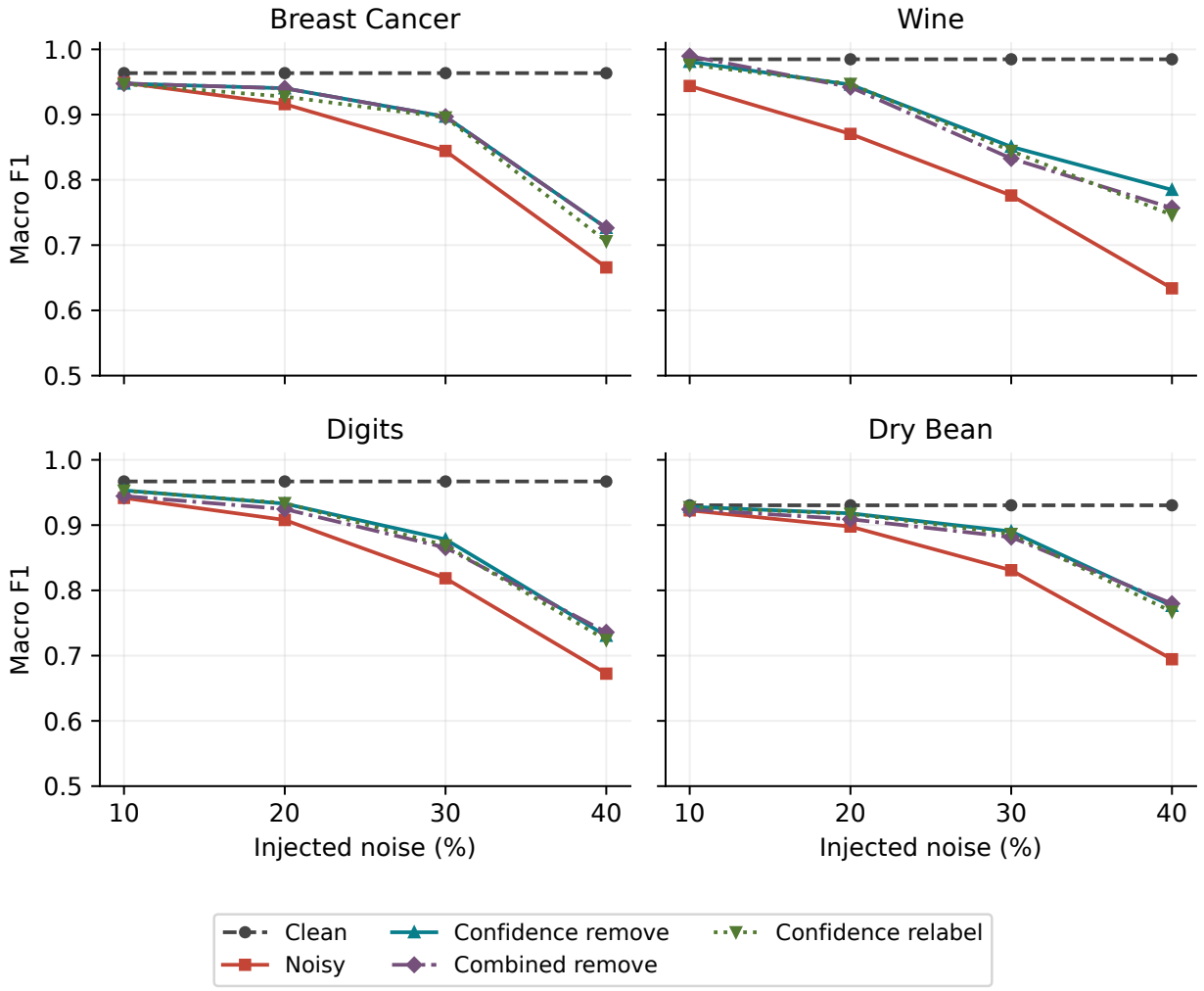



**Figure A.1:** Macro F1 under cyclic class-dependent corruption. Each point averages three models and four seeds within a dataset. The plotted methods and rates match the main accuracy comparison.

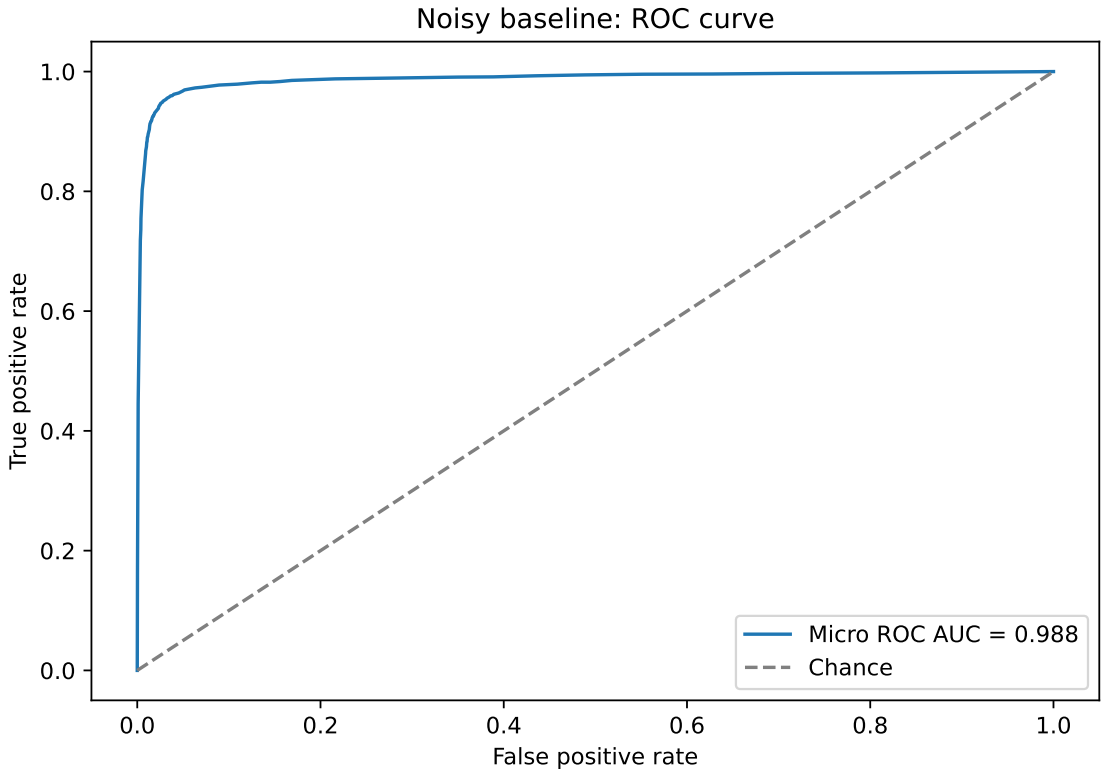



**Figure A.2:** Micro-averaged ROC curve for the noisy Dry Bean RF classifier, symmetric 20% corruption and seed 42. Micro ROC AUC is approximately 0.9878; the main metric for this model is macro one-vs-rest AUROC, approximately 0.9874.

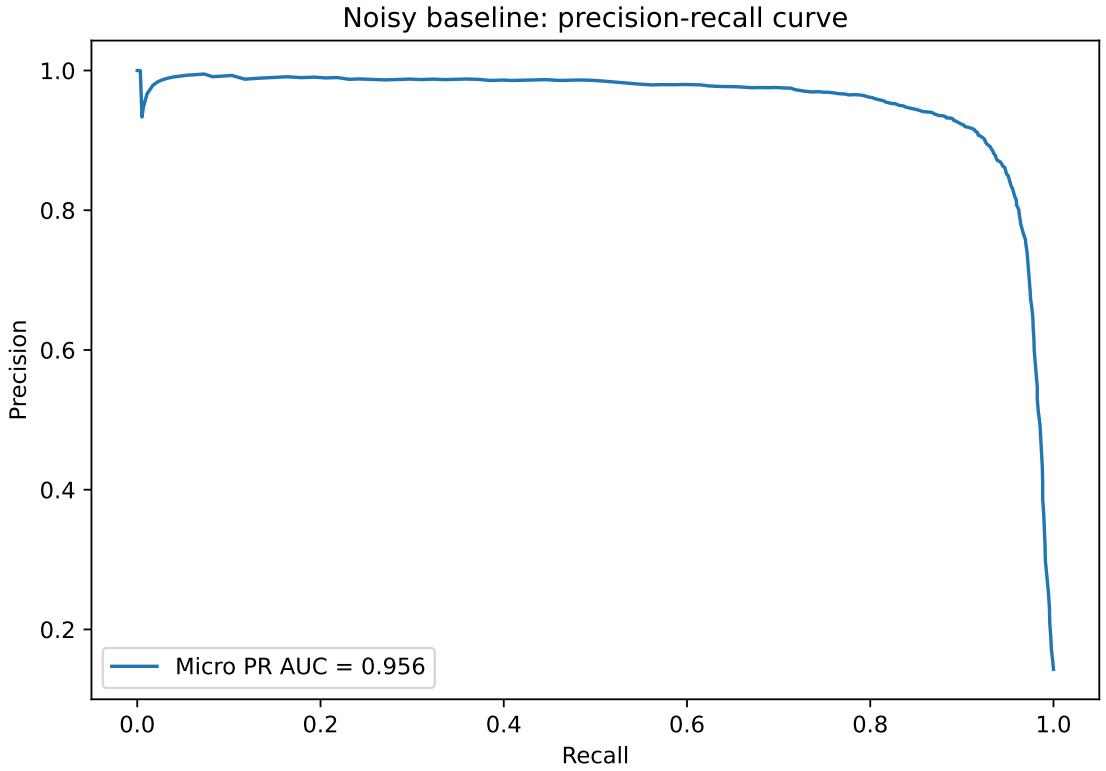



**Figure A.3:** Micro-averaged precision–recall curve for the same noisy classifier. Trapezoidal PR AUC is approximately 0.9558. This is not average precision.